

ASE-project

LifeSphere AI: AI-Powered Adaptive Personal Life Management Platform

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Problem Statement

Managing important personal documents, digital assets, subscriptions, warranties, and emergency information has become increasingly challenging due to the use of multiple applications and physical records. People often forget document renewal dates, lose important files, miss subscription payments, and face difficulties accessing essential information during emergencies. Existing document management systems mainly provide storage facilities but lack intelligent assistance, automation, and adaptive decision-making. Therefore, there is a need for an AI-powered platform that securely manages personal information, provides smart document analysis, intelligent reminders, and adaptive access control to simplify everyday life management.

Abstract

Managing personal documents and digital assets efficiently is a growing challenge in today's digital world. This project proposes **LifeSphere AI**, an AI-Powered Adaptive Personal Life Management Platform that securely stores and manages important documents such as Aadhaar, PAN, Passport, Driving License, Insurance, Medical Records, and other personal information in a centralized system. The platform utilizes **Artificial Intelligence (AI)** and **Optical Character Recognition (OCR)** to automatically extract document details, classify uploaded documents, and provide document summaries. An AI-powered chatbot allows users to search documents, retrieve information, and answer queries using natural language. The system also generates intelligent reminders for document expiry, subscriptions, warranties, and insurance renewals. Adaptive features such as emergency access for authorized nominees, dynamic notifications, and personalized dashboards improve usability and security. The application follows Agile Software Engineering principles and integrates DevOps, DevSecOps, and cloud-native technologies for secure deployment and monitoring. The

proposed system aims to provide a secure, intelligent, and user-friendly solution for managing personal life records through modern AI technologies.

Modern Tools & Technologies Used

Frontend

- React.js
- Tailwind CSS
- HTML5, CSS3, JavaScript

Backend

- Spring Boot
- Spring Security
- REST APIs

Database

- MySQL

AI Technologies

- Google Gemini API
- Tesseract OCR
- LangChain (Optional)

Authentication & Security

- JWT Authentication
- BCrypt Password Encryption
- Role-Based Access Control (RBAC)

DevOps & Cloud

- Git & GitHub
- GitHub Actions (CI/CD)
- Docker

- Kubernetes (Minikube)
- AWS EC2 / AWS S3 (or GCP Free Tier)

Monitoring

- Prometheus
- Grafana

Development Tools

- IntelliJ IDEA
- Visual Studio Code
- Postman
- Maven

Approach

1. The user registers and securely logs into the system.
2. Users upload personal documents such as Aadhaar, PAN, Passport, Driving License, Insurance, and Medical Records.
3. Tesseract OCR extracts important details (Name, Document Number, Expiry Date, etc.) from uploaded documents.
4. Google Gemini AI categorizes documents, summarizes content, and answers user queries through an AI chatbot.
5. The system securely stores all extracted information in the MySQL database.
6. Intelligent reminders are automatically generated for document expiry, warranties, insurance, and subscriptions.
7. During emergencies, authorized nominees receive temporary access to shared documents through adaptive role-based access control.
8. Administrators monitor user activities, audit logs, reports, and system health through an admin dashboard.
9. The application is containerized using Docker, deployed on Kubernetes, and monitored using Prometheus and Grafana for performance and security.